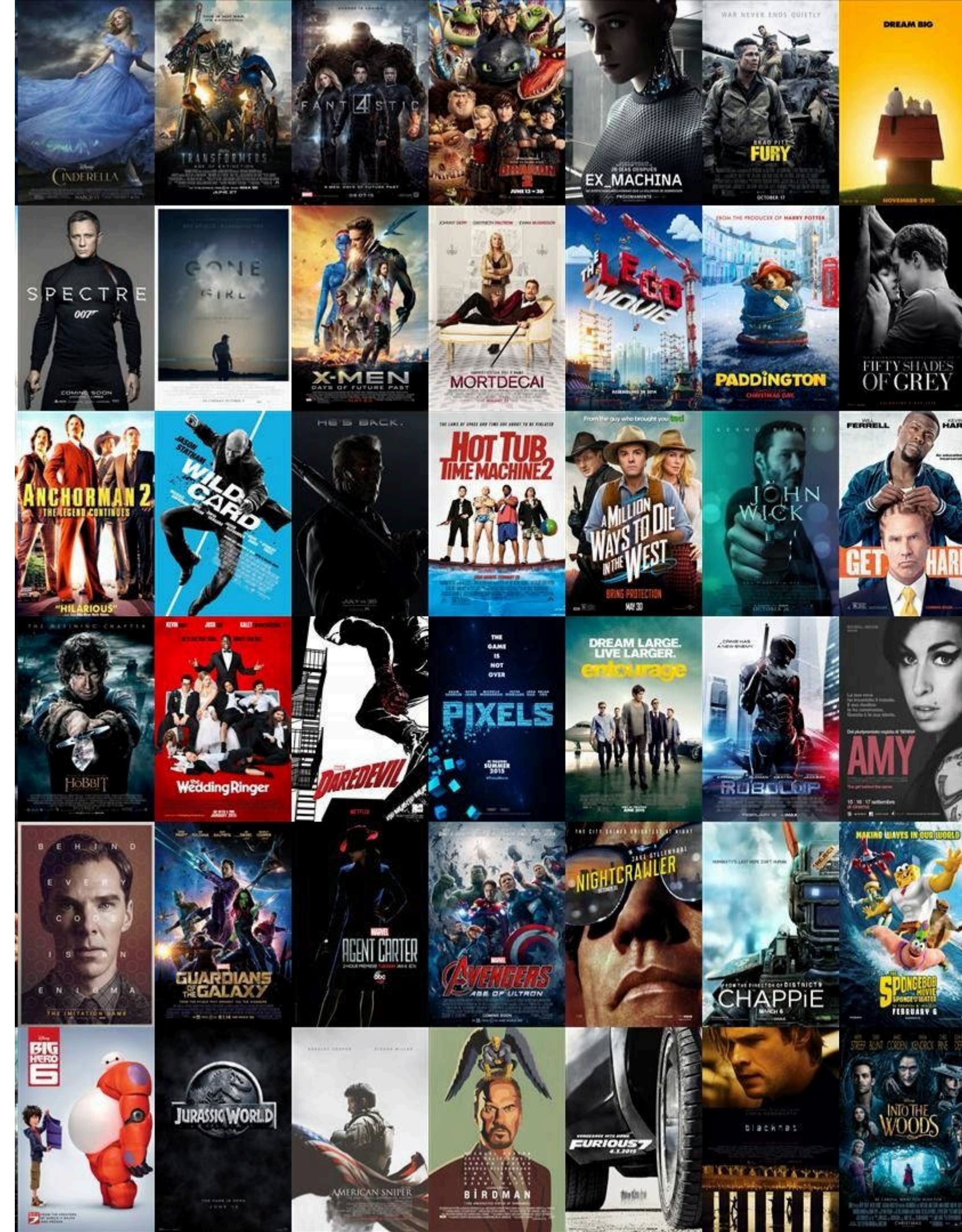


Pre-Release Film Revenue Prediction

Can ML predict box office success before a film releases?

Iciar • Lara • Laura • Olivia • Ryann • Guo Yuting



Index

1. Problem & Scientific Question
2. Dataset & Cleaning
3. Leakage Prevention
4. Multimodal Feature Engineering
5. Model Pipeline & Ensembles
6. Results & Interpretability
7. Limitations & Future Work



Problem and background



CENTRAL RESEARCH QUESTION:

Does combining structured metadata, synopsis text and poster visuals beat any single modality alone?

PREDICTION TARGETS:

Regression

$y = \log_{1p}(\text{revenue}) \rightarrow \text{continuous output}$

Classification

profitable = 1 if revenue $\rightarrow$ $1.5 \times$ budget

Problem and background

WHY IS REVENUE PREDICTION HARD?

Range spans 10+ orders of magnitude

→ log transform essential

Missing & placeholder data (budget=0 $\approx$ unknown)

→ strict cleaning

Strong temporal distribution shift

→ chronological split, not random

Post-release leakage (votes, popularity)

→ explicitly excluded

Sequels & franchises dominate top grossers

→ is_sequel flag

Why Multimodal Learning?

Single Modalities are Incomplete



Combining Modalities
Outperforms Any Single
Representation

Modality	Representation
Metadata	Scale & Timing
Synopsis	Theme & Narrative
Poster	Visual Tone
Talent History	Historical Success

Dataset



+



Poster
Synopsis
Metadata

(imdb_id)

Cast &
Directors

Cleaning

1,413,435

Raw TMDB rows

1,363,284

Remove unreleased films

1,223,447

Drop adult content

16,602

Ensure budget & revenue > 0

10,652

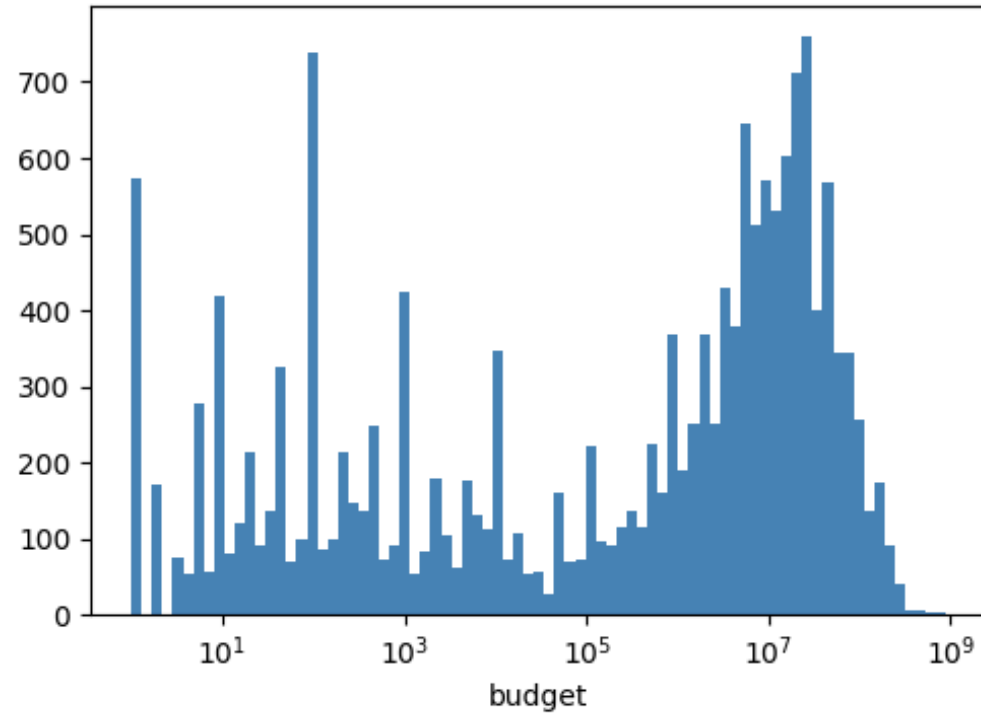
Min \$10k budget filter

9,949

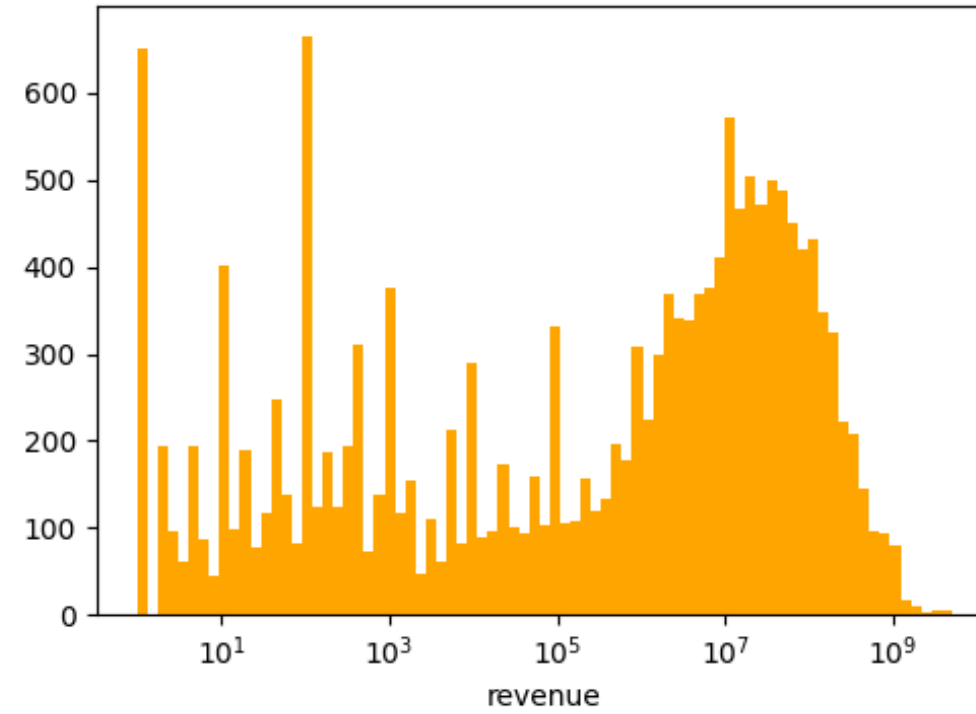
Remove invalid release date + empty overviews

Exploratory Data Analysis

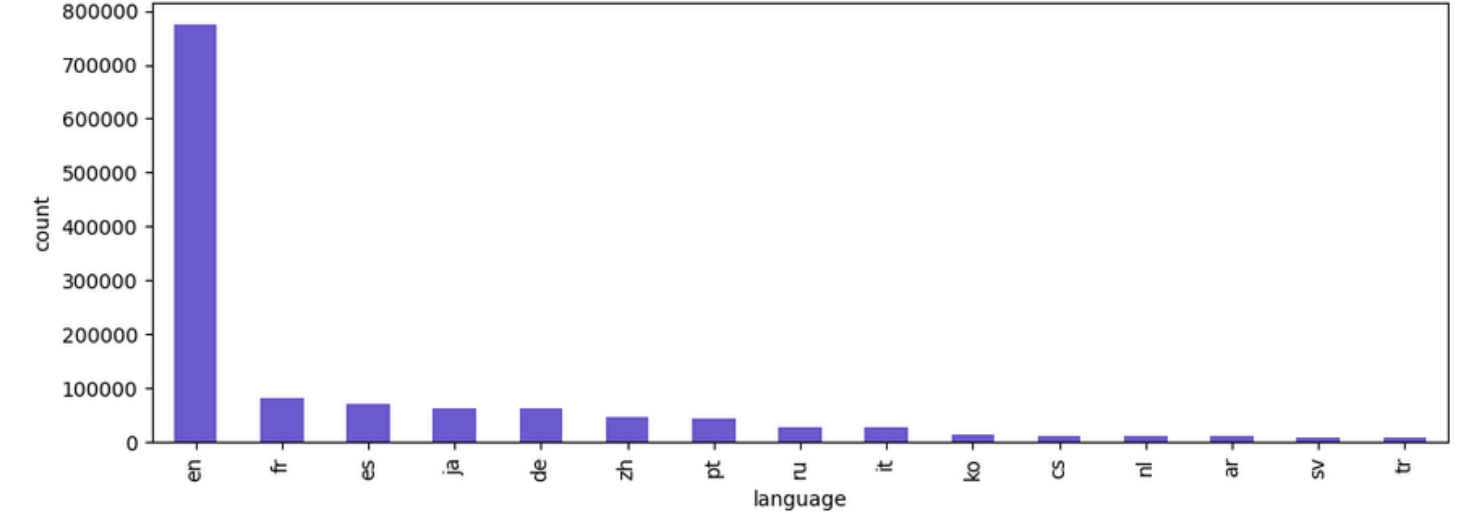
pre-cleaning budget distribution (log scale)



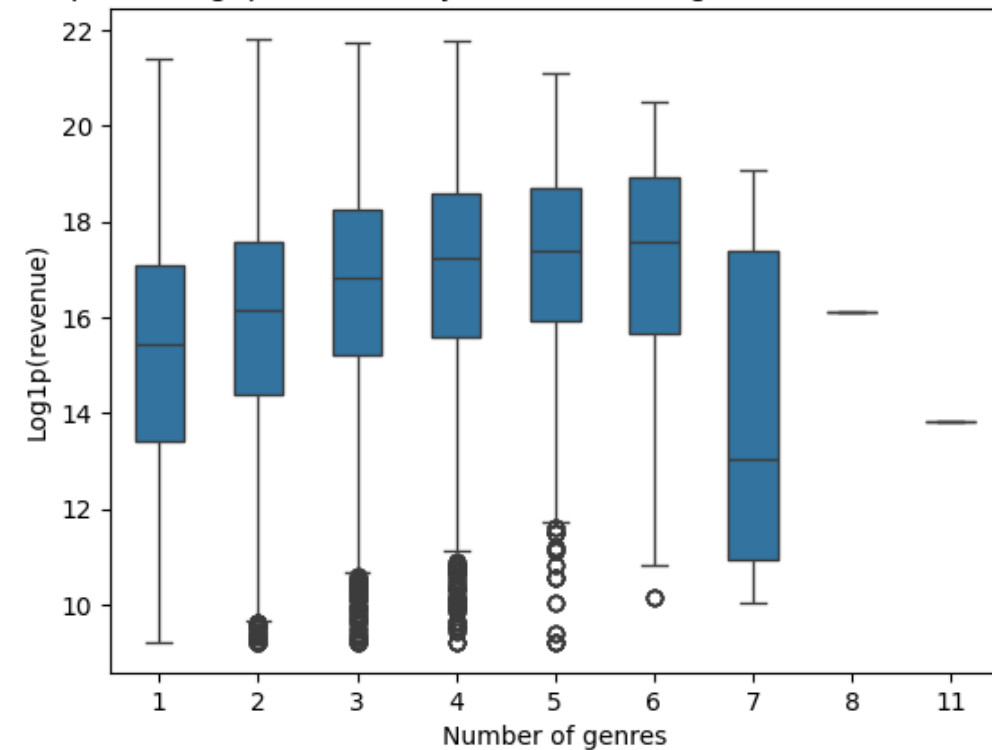
pre-cleaning revenue distribution (log scale)



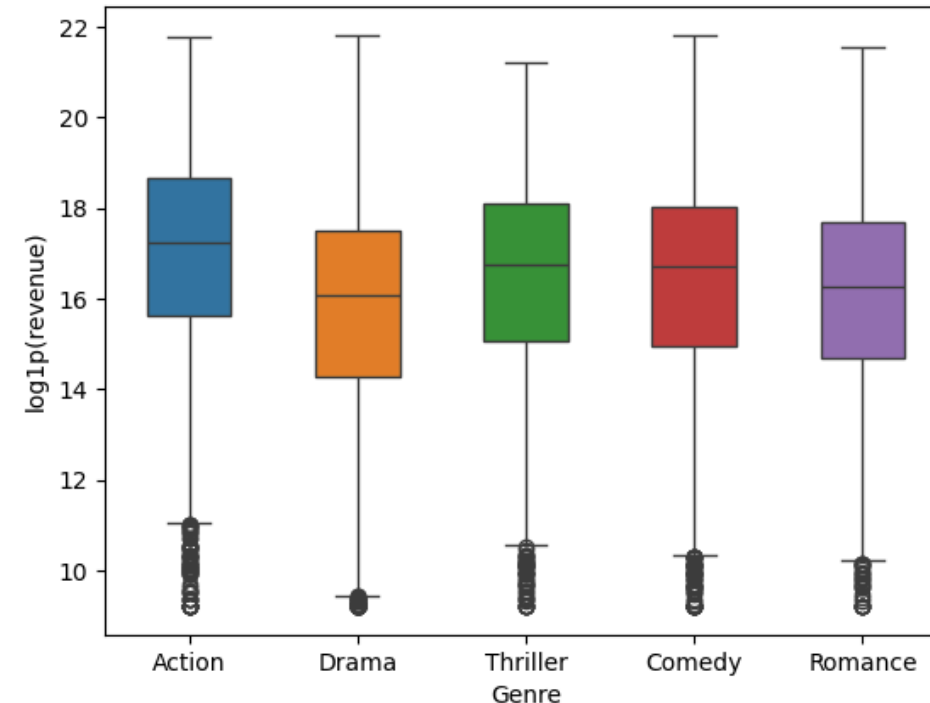
top 15 original languages (pre-cleaning)



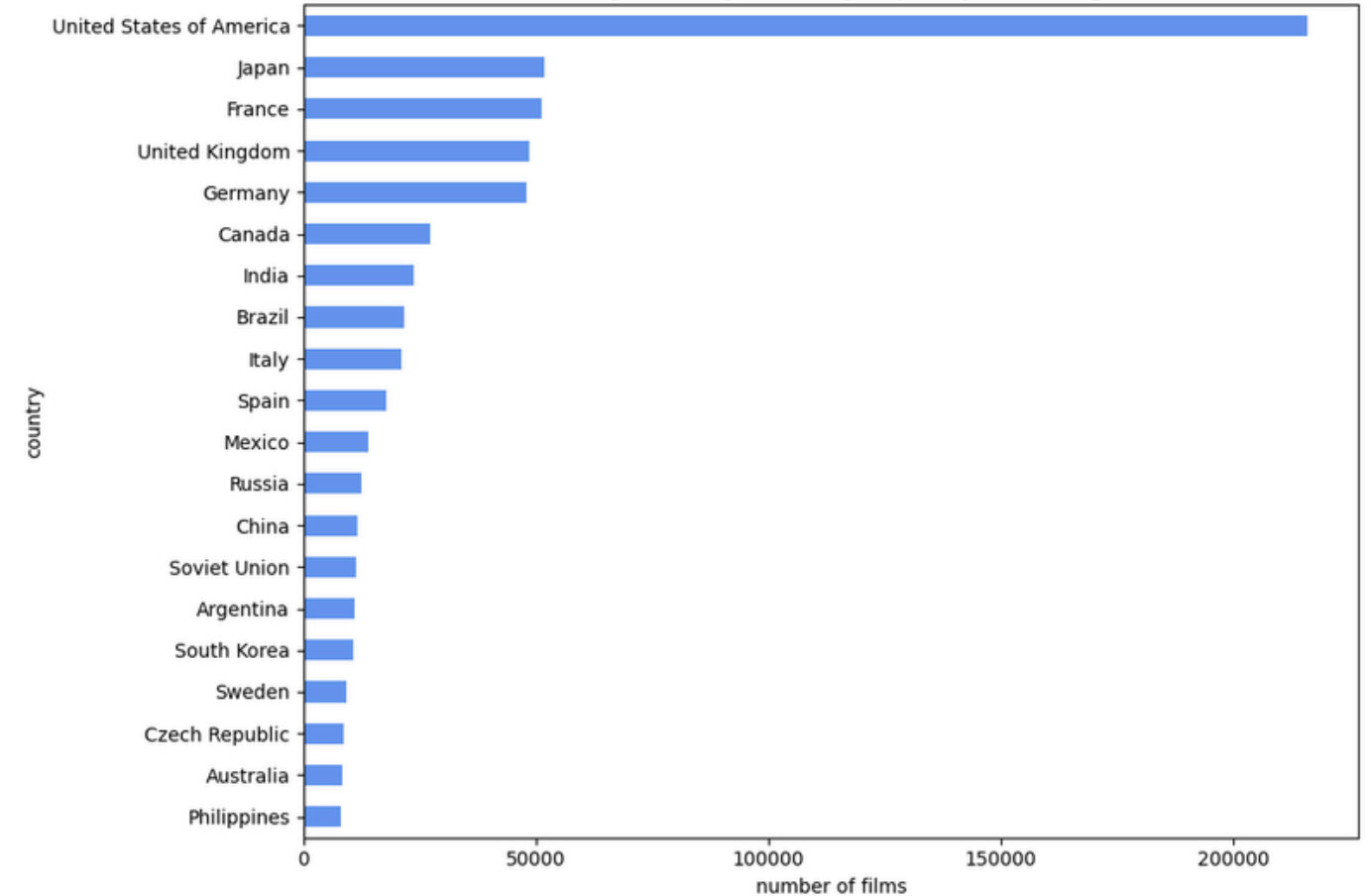
Boxplot of $\text{Log1p}(\text{revenue})$ by the number of genres a movie belongs to.



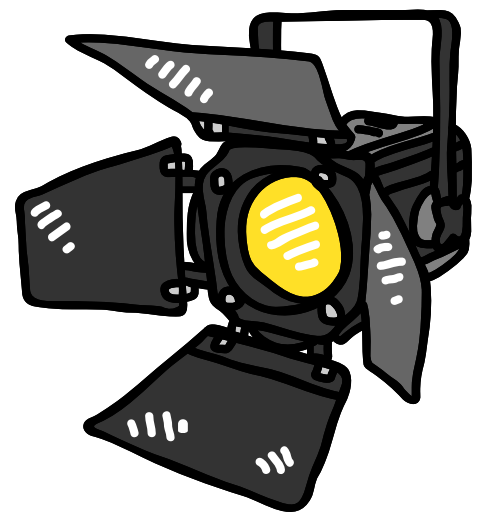
Boxplot of $\text{log1p}(\text{revenue})$ by the top 5 most common genres.



films produced per country (top 20, pre-cleaning)

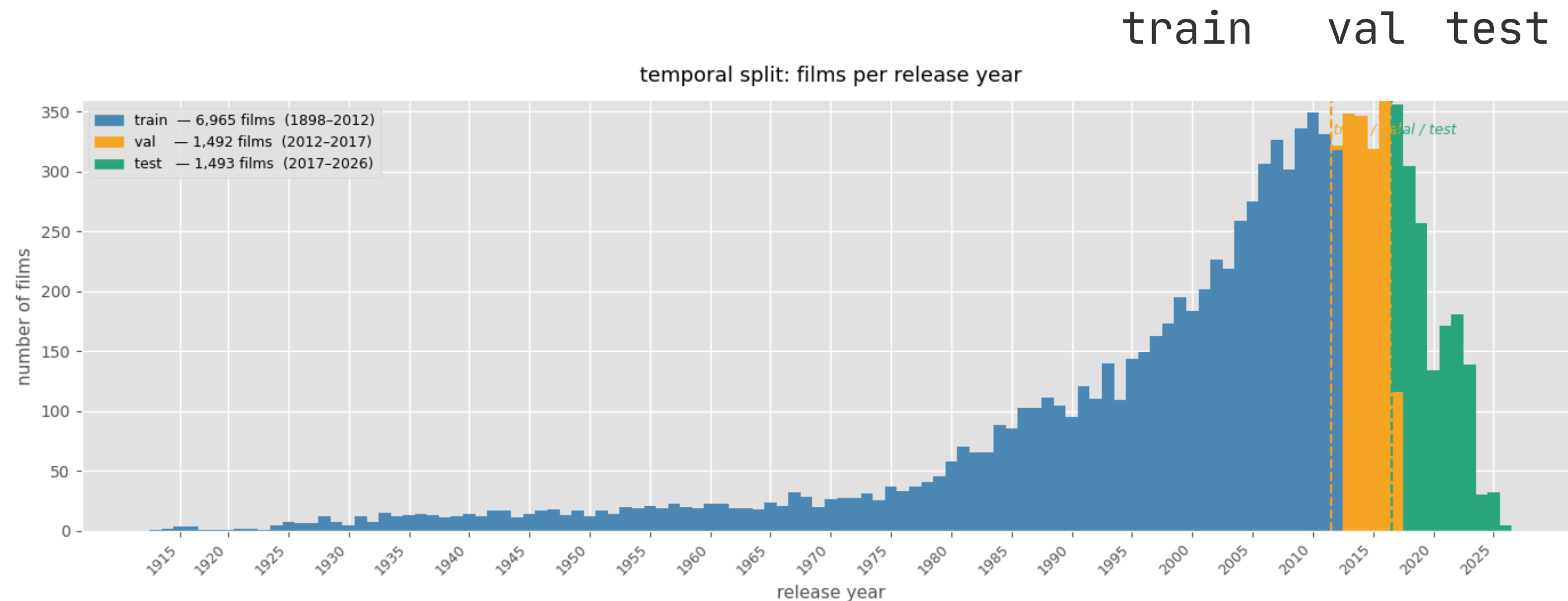


Train/Val/Test Split



70% Training
15% Validation
15% Testing

Why Temporal Splitting Matters
Films must be predicted using only
information available from the past



Data Leakage Analysis



These features inflated performance by revealing **future** audience behavior.

Removed Features:

- `vote\_average`
- `vote\_count`
- `popularity`

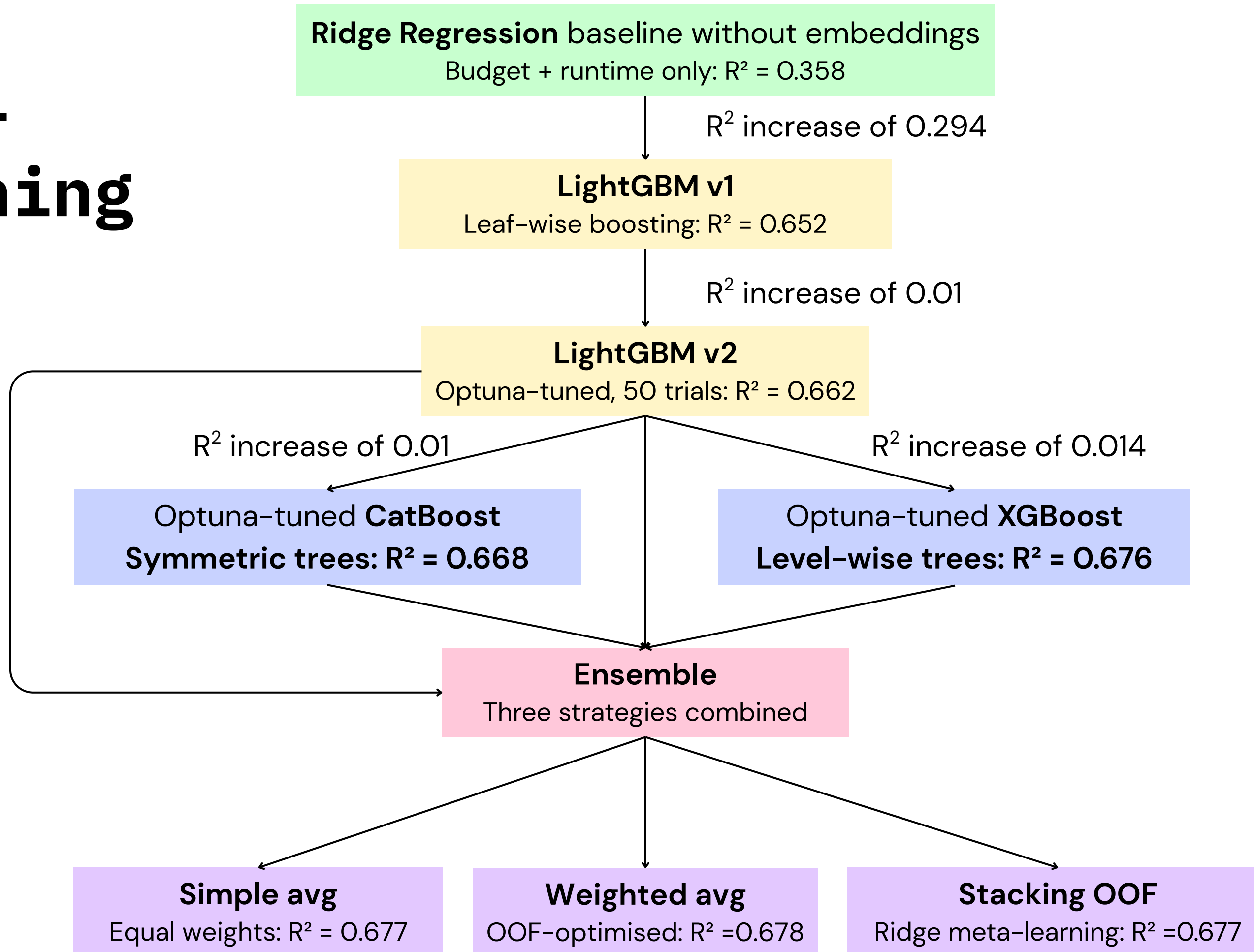
Result

+11.7% artificial R^2 increase when leakage was allowed

Multimodal Feature Engineering

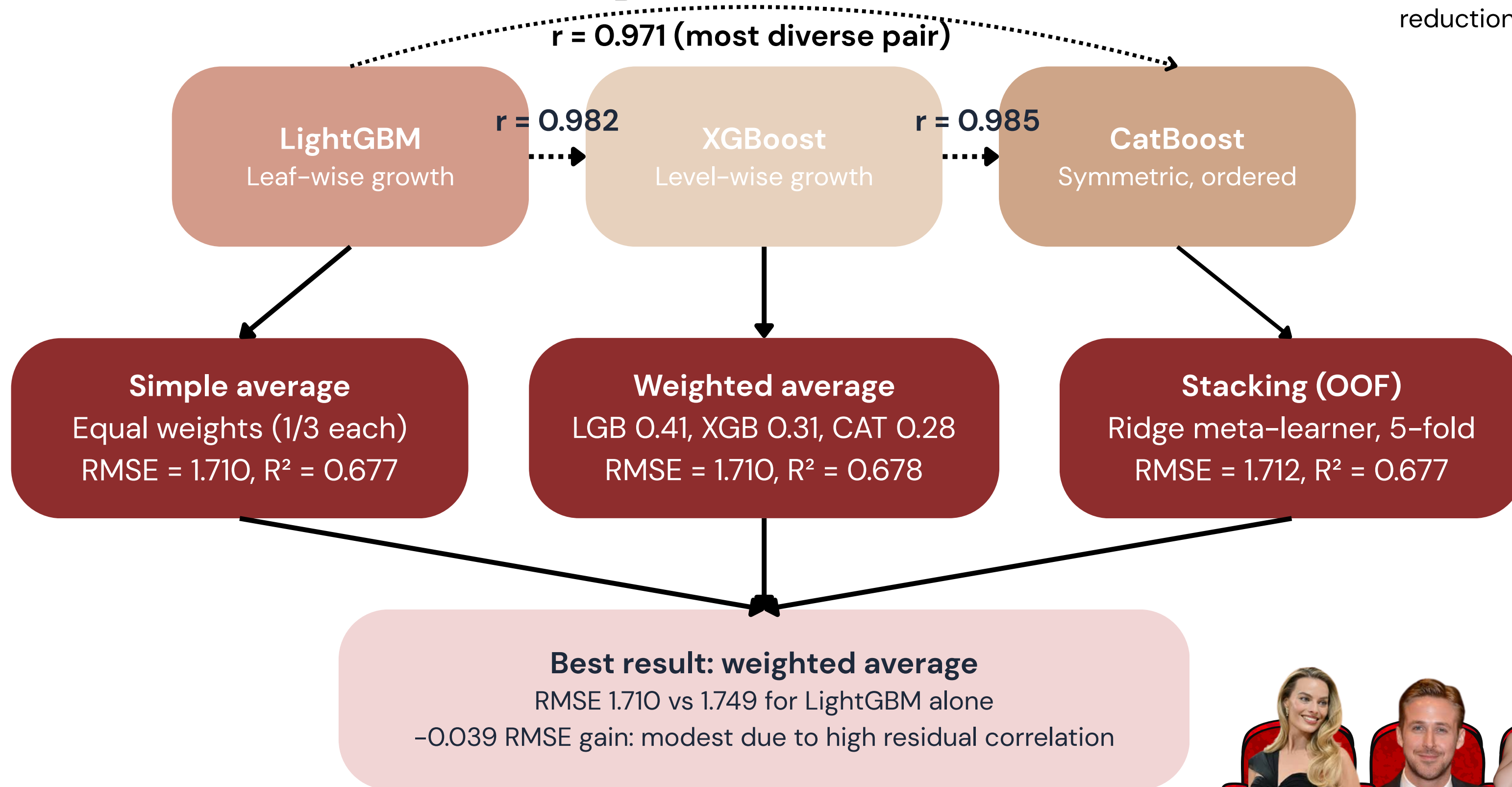
Structured Metadata	Genre OHE	Talent History	Synopsis Embeddings	Poster Embeddings
8 dim	19 dim	2 dim	384 dim	512 dim
•log_budget – log1p transform (right-skew fix) •runtime, release_month, release_year •is_english, is_us_production (binary flags) •num_genres, is_sequel (franchise proxy)	•19 genre dummies from TMDB •Vocab fixed on train → no leakage •Confirms Action = highest revenue	•director_hist_log_rev – expanding median •cast1_hist_log_rev – lead actor history •shift(1) → no target leakage •Cold-start: Bayesian shrinkage prior	•all-MiniLM-L6-v2 sentence transformer •384-dim semantic vector per synopsis •Captures thematic content & tone •Upgraded to mpnet-768 in v2	•CLIP (ViT-B/32) – language-aligned vision •512-dim visual features per poster •Cached → computed once •PCA 512→50 dims in improved model

Model training



Ensemble Strategies

Pairwise residual correlations 0.971–0.985:
high but not perfect: measurable variance
reduction



The gain was limited not by strategy choice, but by the high correlation between three gradient boosting variants.



Ablation Study

How Much Did Each Modality Contribute?

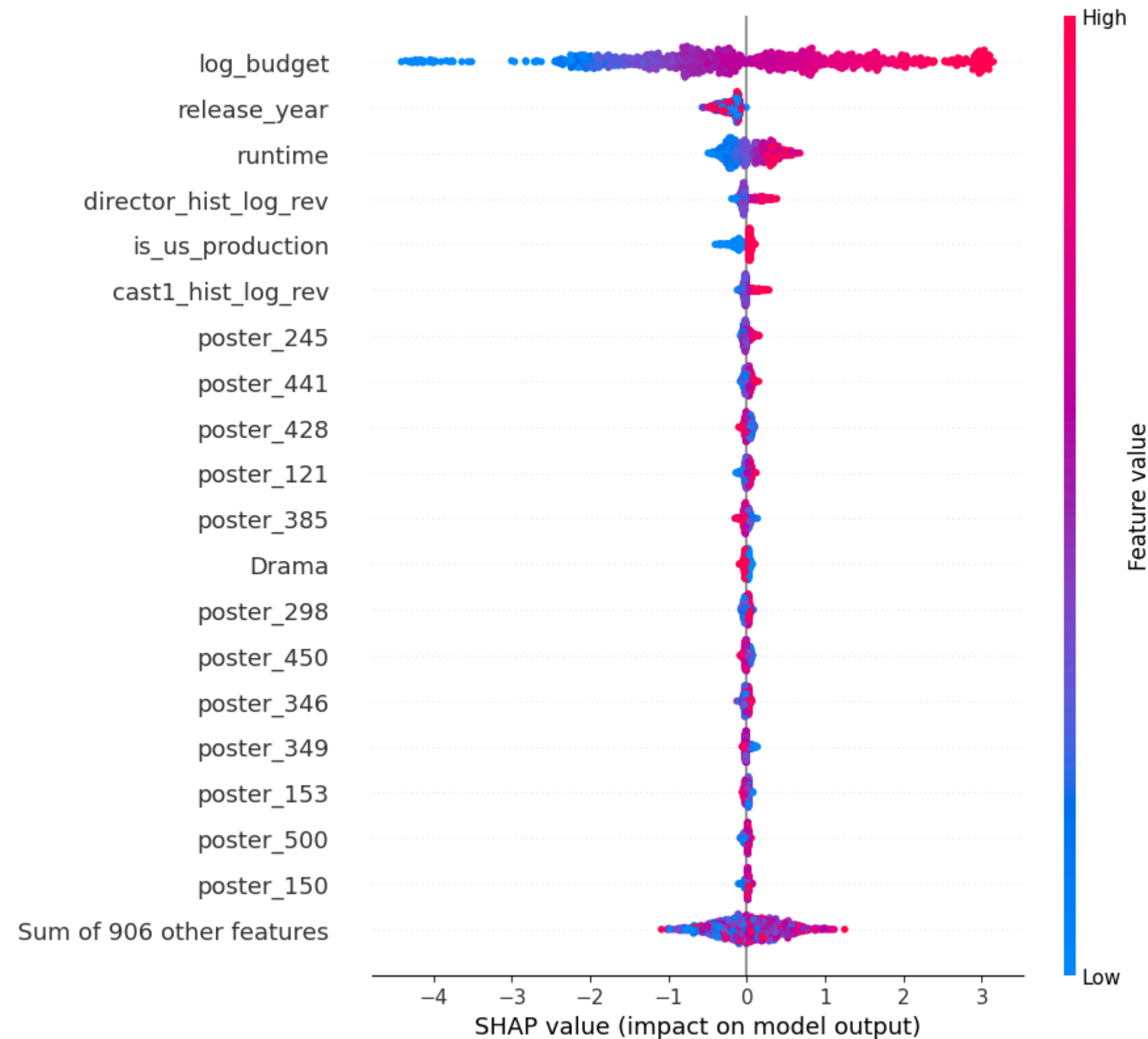
Features Added	R ²
Structured Only	0.617
+Genre	0.624
+Talent	0.630
+Synopsis	0.633
+Poster	0.639

Key Finding

Performance improved
Incrementally as
modalities were added



Shap Interpretability



What Did the Model Learn?

- Budget was the dominant predictor
- Strong director/cast history increased predictions
- US productions consistently ranked higher
- Poster embeddings contributed meaningful signal

Key Insight

The model learned economically plausible patterns, not random correlations

Results and analysis

$R^2 = 0.677$

final ensemble

+31.9 pp R^2

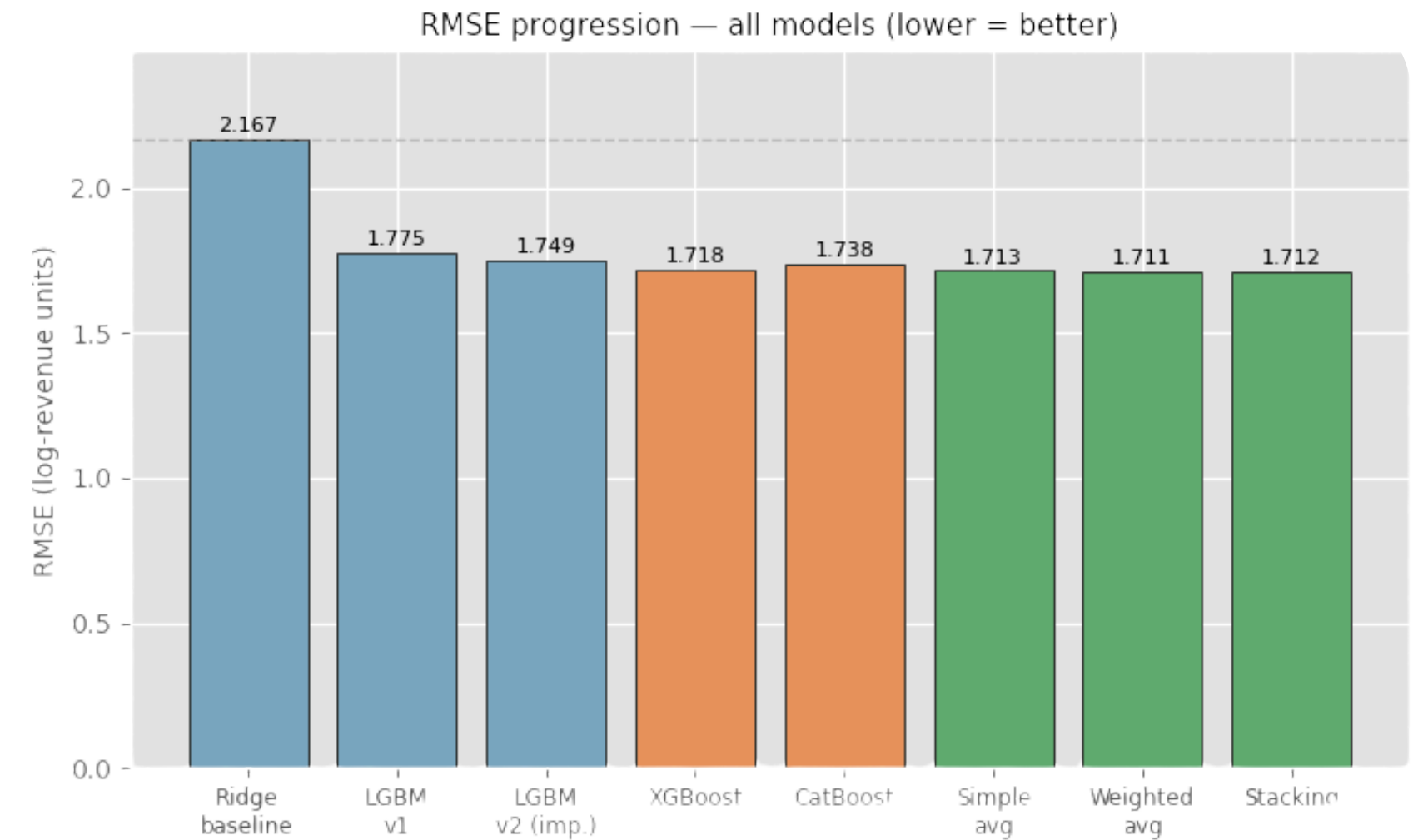
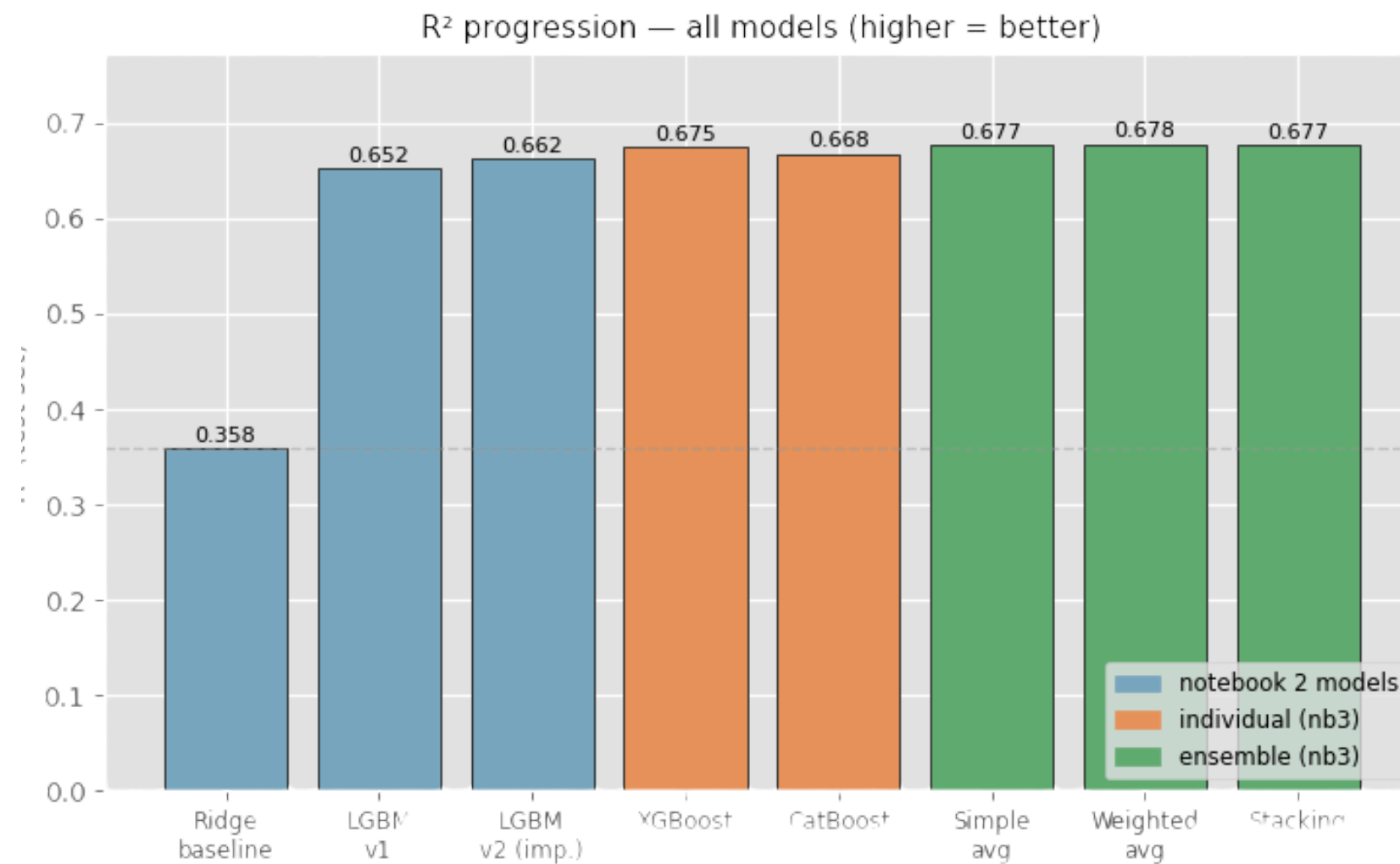
over baseline

1.71 RMSE

log-revenue units

- Published benchmarks 0.50–0.70 R^2 → competitive result
- Model explains 67% of variance → 30% is due to unobservable factors
- Typical error is 5.5x in revenue

Full model comparison



- Biggest increase in performance came from adding multimodal features
- Ensembling is only slightly better than individual models

Ensembling analysis

Model	RMSE	R ²	ΔR ²
Ridge (budget + runtime)	21.672	3.584	0
LightGBM v1 (924 dims)	17.754	6.520	2.936
LightGBM v2 (improved)	17.490	6.622	3.038
XGBoost (tuned)	17.176	6.751	3.167
CatBoost (tuned)	17.375	6.675	3.091
Simple avg ensemble	17.134	6.767	3.183
Weighted avg ensemble	17.110	6.776	3.192
Stacking (OOF Ridge)	17.125	6.770	3.186

3 ensembling strategies
provided nearly identical
results

Why?

1. Similar architecture,
all gradient-boosted
decision trees
2. Models share same input
features
3. Models fail on same
films

Limitations & Challenges

Scientific Limitation:

Box office success still contains irreducible uncertainty.

Limitations:

- Sparse talent history for newer directors
- High-dimensional feature space (925 dims)
- Poster embeddings added weak but noisy signal
- Temporal distribution shift across decades

Challenge:

Switched from ResNet → CLIP for embeddings, took a lot of time.

Conclusion & Future Work

Main Conclusion

Multimodal representation learning improved performance more than model complexity alone.

Future Work

- Expanding dataset size and sources
- Adding marketing spend data
- Improving synopsis and poster embeddings
- Adding a dedicated visual branch (CNN)

Thank You!

Iciar • Lara • Laura • Olivia • Ryann • Guo Yuting